

MSBA Job Matching - Final Technical Summary

Problem and Production Objective

The system supports a graduate business career advisor who must decide which opportunities to forward, hold, or investigate for MSBA international students. It accelerates first-pass review while keeping role fit, incomplete evidence, and work authorization as separate decisions.

Production Architecture

Stage	Production behavior
Route	Distinguish search requests from job postings
Retrieve	Rank 80,000 real postings with constraints and dedupe
Classify	Use a 3.8 MB word+character TF-IDF SVM artifact
Govern	Override missing fields; Hold explicit restrictions
Respond	Return cited evidence and advisor action

Measured Improvement

Model	Training rows	Macro F1	Role
M2 baseline	80,000	0.791	Comparison
M3 PEFT LoRA	2,000	0.873	Adaptation experiment
Final TF-IDF SVM	77,135 deduped	0.984	Production decision layer

On the same 4,000-row subset, macro F1 improves by +0.192 over the M2 comparator. Novel-text macro F1 is 0.979; unseen-company macro F1 is 0.980; newest-20% time-split macro F1 is 0.976.

Why This Model

A training-size curve shows 0.898 macro F1 at 2,000 rows, 0.944 at 8,000, 0.965 at 20,000, and 0.984 at 80,000. The final linear model is more accurate than the current LoRA experiment, trains in under one minute, and is straightforward to reproduce. A larger transformer is not justified until human labels exist.

Real Runs and Failure Handling

Case	Model	System	Decision support
entry level analytics search	N/A (search)	N/A (search)	Advisor review before forwarding
senior engineering posting	low_fit	low_fit	Advisor review before forwarding
authorization restriction posting	high_fit	high_fit	Hold - explicit authorization restriction requires advisor verification
thin metadata posting	medium_fit	unclear	Advisor review before forwarding

Governance and Residual Risk

The saved casebook returns evidence for all four workflows, forces review for one explicit authorization restriction, and contains zero unsupported authorization claims. Exact duplicates are removed before training; novel-text, unseen-company, and time-based slices are reported. A scan of all 785,741 source rows identifies 230 unique title-level authorization candidates. Weak role-fit labels remain a limitation, so advisor fields are never auto-filled.

Business Recommendation

Use the included 1,000-row advisor queue to run a controlled Career Center review: 800 low-margin role-fit cases and 200 authorization candidates. Measure high-fit precision, minutes saved per posting, override rate, and authorization-evidence accuracy while keeping the system as decision support. The queue and annotation guide make that review reproducible without fabricating human labels.

Production Assets

The repository contains reproducible training, random and time-based evaluation, a full-source authorization audit, annotation guide and queue, casebook, CLI, tests, model artifact, M2-M4 notebooks, governance documentation, slides, and a demonstration workflow.